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Elements of the Archean thermal history and apparent polar wander of the eastern Kaapvaal Craton, Swaziland, from single grain dating and paleomagnetism

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Single grains of zircon, hornblende, biotite and feldspar have been dated to define the thermal history of the Archean Mbabane Pluton, Swaziland. Coincident $^{207}Pb/^{206}Pb$ zircon and $^{40}Ar/^{39}Ar$ hornblende ages suggest that the pluton underwent rapid cooling to about 450 °C at the time of emplacement at ~ 2690 Ma. Because the Mbabane Pluton is one of a suite of granites which represent the last major Archean intrusive event in the eastern Kaapvaal Craton, this time of emplacement marks the end of cratonization. Much younger biotite and feldspar ages indicate argon loss, perhaps due to several low-temperature events related to dike intrusion at 2000–2300 Ma. From such a thermal history, the age of magnetization of a paleomagnetic pole (pole latitude = 19.7 °N, pole longitude = 105.7 °E, $A_{95} = 9.4°$, $K = 67$) obtained from 5 sites can be determined. Based on the unblocking temperature of the magnetic carrier and the consistent direction seen in several sites, it appears that the hornblende age (and coincidently the $^{207}Pb/^{206}Pb$ zircon age) most accurately represents this age of magnetization. Thus, the Mbabane Pluton yields a precisely dated Archean (2687 ± 6 Ma (2σ), hornblende plateau age) paleomagnetic pole which is different than other Archean poles such as that from the nearby Usushwana Complex (2875 ± 40 Ma). Although the two intrusions are separated by less than 12 km and there is no evidence of relative structural rotation between them, the paleomagnetic poles from the two bodies are 65 ° apart. This implies that there was significant apparent polar wander during the late Archean.

1. Introduction

Recent developments in micro-dating techniques are revolutionizing the field of geochronology. Rather than age determinations being made on bulk samples of zircon (for the U-Pb system) or hornblende, biotite and feldspar (for the argon system), individual grains or regions in a grain can now be dated with great precision (for example, [1–4] for the U-Pb system, [5–9] for the argon system). Studies using these micro-techniques have shown that, in some cases, results from bulk samples may be meaningless due to the mixing of several components of different ages [10,11]. The new single grain techniques can eliminate this contamination problem because each grain to be dated can be carefully selected. In addition, the procedure for separating grains is simplified because elaborate separation techniques are no longer needed to produce the several milligrams of pure material formerly required. Single grain dating methods in $^{40}Ar/^{39}Ar$ and U-Pb, then, offer simple, fast, and precise means of dating minerals.

Ages determined from different minerals and isotopic systems can provide valuable information for the determination of the thermal history of a pluton [12]. Such a detailed history is far more useful than an age on a single mineral or on a whole-rock sample alone. Although the exact blocking temperatures for the various minerals are not precisely known, it is generally recognised that zircons block (or close) to diffusion of lead at temperatures in excess of 700 °C, hornblende to argon diffusion between 450 °C and 550 °C (and perhaps higher), biotites to argon diffusion between 200 °C and 350 °C, and feldspars to argon diffusion at temperatures usually less than 200 °C [12–14]. Thus, with these four minerals spanning the blocking temperature scale from about 200 ° to 700 °C or more, magnetization ages for all

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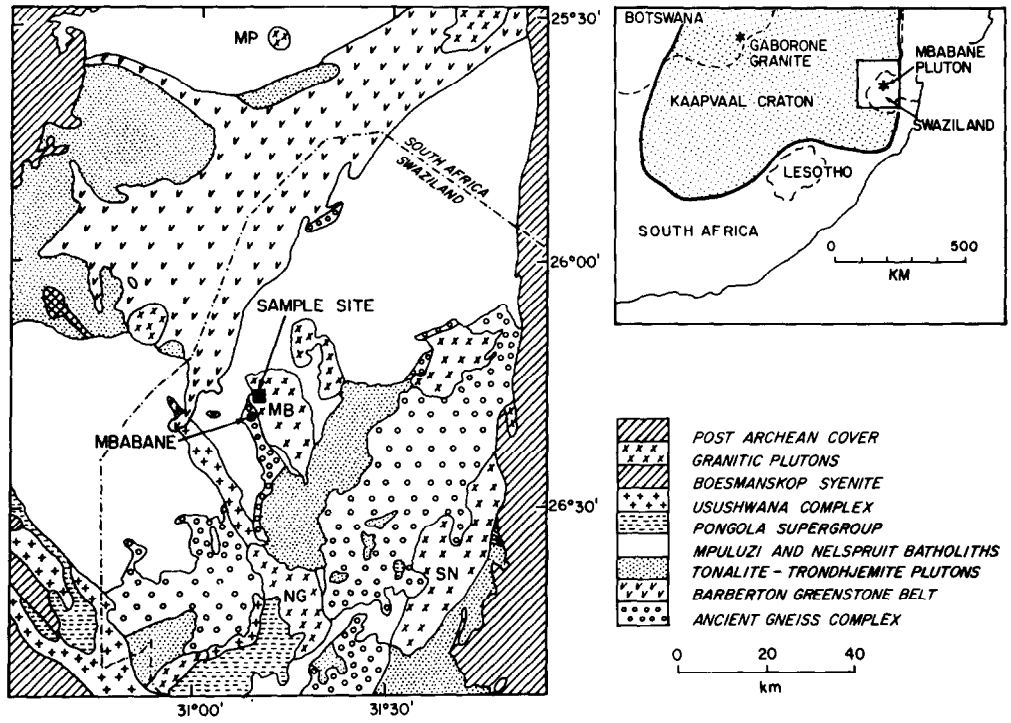


Fig. 1. Inset: Sketch map of southern Africa showing the Kaapvaal Craton and the locations of the Gaborone Granite, Botswana and the Mbabane Pluton, Swaziland. The area enlarged in the main map is shown by the box. Main map: Generalized geologic map of the eastern Kaapvaal Craton adapted from the 1:250,000 geologic map of Swaziland showing the location of the younger granite plutons. *MB* = Mbabane Pluton, *MP* = Mpageni Pluton, *NG* = Ngwempisi Pluton, *SN* = Sinceni Pluton. The sampling site location in the Mbabane Pluton is shown as is the city of Mbabane.

magnetic minerals relevant to paleomagnetic work can be determined. This thermochronometricpaleomagnetic approach developed in detail originally with bulk samples [12,15], is critical to the development of an understanding of the thermal evolution of Archean cratons and their relative motions.

In this study, the single grain zircon evaporation technique of Kober [3,4] and the single grain $^{40}Ar/^{39}Ar$ laser step heating technique outlined by Layer et al. [6] are used to estimate the age of magnetization of the Mbabane granite pluton located in north-central Swaziland, southern Africa. This pluton was chosen for study because, with an approximate Rb-Sr age of 2500–2700 Ma [16,17], it represents one of the last intrusive events in one of the best preserved Archean terranes on Earth. Following the intrusion of the Mbabane and other related plutons, such as the Mpageni, Ngwempisi and Sinceni (Fig. 1), the area of eastern Kaapvaal craton has had a very quiet tectonic and thermal history [17] with the exception of an episode of dike emplacement between 2000 and 2300 Ma [18] and eruption of Karoo lavas in the Mesozoic. Previous paleomagnetic work in north-central Swaziland had yielded a 2875 ± 40 Ma (Rb-Sr and Sm-Nd isochron ages) pole from the mafic-ultramafic Usushwana Complex [19]. We compare the pole from the Mbabane Pluton with that from the Usushwana Complex in an attempt to identify the existence of apparent polar wander in the late Archean in southern Africa.

2. Geologic summary

The Mbabane Pluton is a coarse-grained porphyritic granitic pluton of high topographic relief located in north-central Swaziland east of Mbabane (Fig. 1). The pluton is exposed as a tear-drop shaped structure 35 km long with a maximum width of 18 km. Medium-grained biotite-rich inclusions are common in this pluton and

appear to be disc-shaped and randomly oriented [20]. The pluton intrudes gneisses and migmatites of the Ancient Gneiss Complex in the south, and homogeneous massive adamellites of the Mpuluzi Batholith in the north [17] (Fig. 1). Xenoliths of these older materials are found in the pluton [21].

The Mbabane Pluton is cut by a few dolerite dikes of presumed Proterozoic age. Their frequency of occurrence is much lower than in the surrounding Mpuluzi Batholith, perhaps implying that the Mbabane Pluton was not as fractured or jointed as the older batholith and did not allow for easy intrusion of dike material at that time [21]. The dikes seen in the pluton are all vertical or subvertical, a criterion used to infer that the pluton has not been tilted significantly since their intrusion. Unfortunately, this, and the lack of any visible faults in the area are the only evidence that is available regarding a structural correction, and, for the following discussion, it is assumed that no tilting of the pluton has occurred.

Seventy samples were collected at seven sites (MBA–MBG; 10 samples per site) in the Mbabane Pluton for paleomagnetic and geochronologic work. Each sample consists of a 6–10 cm long field-drilled core 2.5 cm in diameter. All samples were oriented in the field using both a sun compass and a magnetic compass. The sites are located four kilometers northeast of Mbabane along the flank of Sibebe Hill (26°16′S, 31°10′E, Fig. 1). Here the pluton is exposed as a dome with vertical exposure of over 500 m and shows only a thin (less than one cm) weathering rind. Six sites were located near the base of this dome to minimize possible magnetic overprinting due to lightning strikes. Site MBE was taken from a stream pavement in the Mbuluzi River at the base of the dome.

3. Previous age determinations

Samples for Rb-Sr dating were collected from the Mbabane Pluton near Mbabane (close to the area chosen for this study) by Allsopp et al. [16]. Two whole-rock model ages and one feldspar model age were in all agreement and gave a mean age of 2495 ± 60 Ma (recalculated [22] using the new decay constants [23]). Allsopp et al. [16] reported the presence of chloritized biotite, and of strain extinction in quartz grains which may be indicative of alteration or some post-cooling tectonic event. Thus, this model age may not represent either the age of pluton formation or thermal overprinting.

4. U-Pb study

Single grains of zircon, hornblende, biotite, and feldspar have been separated from specimen cores from two sites (MBE and MBC) which show stable paleomagnetic directions. Zircons from these two sites were dated by the single grain evaporation technique recently by Kober [3,4]. In this method, chemically untreated zircons about 80–100 μm in size and embedded in the evaporation filament of a rhenium double filament arrangement, and Pb is then evaporated in the mass spectrometer during heating of the grain. We

TABLE 1

Isotopic data from single zircon evaporation

Sample name	Grain	Mass scans [a]	Evaporation temperature (°C)	Mean $^{207}Pb/^{206}Pb$ ratio [b] and standard error	$^{207}Pb/^{206}Pb$ age (Ma) and standard error
MBE	1	60	1680	0.18418 ± 25	2691 ± 2
MBE	2	62	1670	0.18412 ± 22	2690 ± 2
MBE	1 + 2	122		0.18415 ± 23	2691 ± 2
MBC	1	82	1660	0.18421 ± 26	2691 ± 2
MBC	2	54	1670	0.18422 ± 32	2691 ± 3
MBC	1 + 2	136		0.18422 ± 27	2691 ± 2

[a] Number of $^{207}Pb/^{206}Pb$ ratios evaluated for age assessment.

[b] Observed ratio; non-radiogenic Pb negligible ($^{206}Pb/^{204}Pb > 45{,}000$), errors based on uncertainties in counting statistics. Decay constants are from [23].

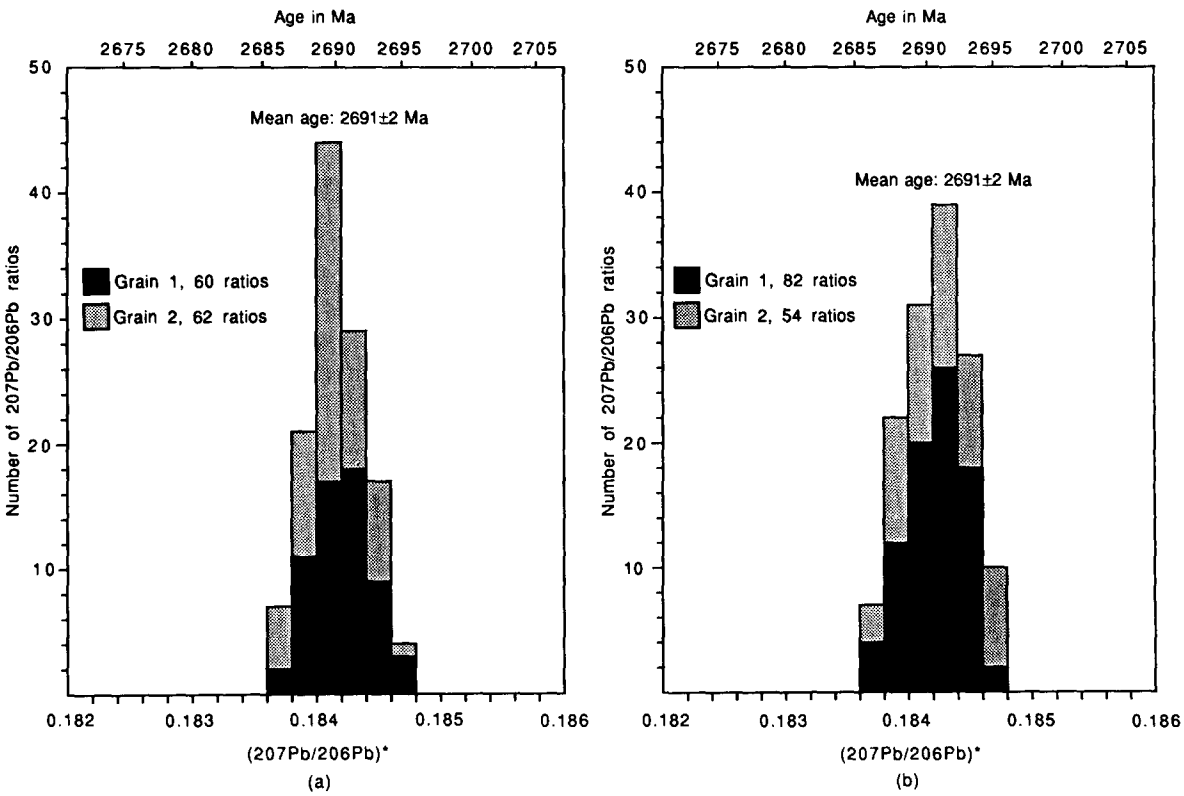


Fig. 2. Histograms showing the distribution of radiogenic lead isotopes derived from evaporation of zircons from the Mbabane Pluton, Swaziland. (a) Site MBE, combined spectrum of 122 ratios from two grains, (b) Site MBC, combined spectrum of 136 ratios from two grains. Mean ages are given with standard errors.

have employed step-wise evaporation involving deposition of radiogenic Pb on the cold ionization filament during heating to about 1660–1680 °C. Kober [3] has shown that the Pb components with the highest activation energy normally reside in the undamaged crystalline zircon phase that shows no post-crystallization Pb loss and therefore yields concordant $^{207}Pb/^{206}Pb$ ages. The evaporation technique has been tested on isotope standards and natural zircons previously dated conventionally and by ion microprobe (see [24] for details), and the results are identical within error, even for zircons with complex metamorphic histories. The technique has proven particularly useful in dating Archean zircons from the Ancient Gneiss Complex of Swaziland [24] and the adjacent Barberton Greenstone Belt [25]. Isotopic measurements were carried out on a Finnigan-MAT 261 mass spectrometer in the Max-Planck-Institut für Chemie in Mainz, and analytical details are reported in [25].

Two zircons each were evaporated from samples from sites MBE and MBC, and the data are presented in Table 1 and Fig. 2. The zircons are clear to yellow-brown in color, free of inclusions and have idiomorphic, long-prismatic shapes. The $^{207}Pb/^{206}Pb$ ratios from both samples define normal distributions and yield identical mean ages of 2691 ± 4 Ma (2σ).

5. $^{40}Ar/^{39}Ar$ study

Single hornblende, biotite and feldspar grains were dated by $^{40}Ar/^{39}Ar$ step heating (Table 2). Each grain was located in a 2 mm diameter hole on an aluminum tray 18 mm in diameter. Each tray held 13 Mbabane grains and 3 grains of a standard mineral, hornblende 3gr, which has been shown to be an excellent single grain standard [6]. Samples were irradiated at the McMaster University reactor for 192 mwh. Gas fractions released

TABLE 2

$^{40}Ar/^{39}Ar$ ages from single grains

Sample name	Grain	Number of fractions [a]	^{40}Ar * volume [b] ($\times 10^{-9}$ ml STP)	Ca/K [c] and standard error	Integrated age (Ma) and standard error
Hornblende					
MBE	1	14	6.9	6.919 ± 0.013	2646 ± 4
	2	16	19.2	6.398 ± 0.007	2666 ± 3
MBC	1	8	6.8	6.240 ± 0.013	2663 ± 4
	2	15	21.3	8.931 ± 0.012	2730 ± 3
Composite [d]		14	54.2	7.410 ± 0.006	2688 ± 3
Biotite					
MBE	1	15	3.8	0.032 ± 0.002	2140 ± 4
MBC	1	15	9.5	0.234 ± 0.001	2088 ± 3
*	1	19	42.9	0.040 ± 0.000(4)	2261 ± 3
*	2	25	94.8	0.050 ± 0.000(3)	2155 ± 3
*	3	25	19.9	0.212 ± 0.014	2177 ± 3
Feldspar					
MBE	1	11	3.8	1.612 ± 0.004	1671 ± 3
	2	10	2.1	1.035 ± 0.004	1230 ± 3

[a] Number of gas fractions measured during step heating up to sample fusion.

[b] Radiometric ^{40}Ar volume of the sample.

[c] Ca/K is calculated from $^{37}Ar_{Ca}/^{39}Ar_{K}$.

[d] The composite spectrum is made up of a weighted sum of the four individual spectra, "repartitioned" with a synthetic temperature schedule.

* These three biotites are from a crush of material from sites MBB, MBC and MBE.

Decay constants are from [23]. The standard used is hb3gr (1071 Ma). $J = 0.02303 \pm 0.00004$ for all samples except those marked with * where $J = 0.02614 \pm 0.00006$.

by laser heating of single grains were analysed on a VG Micromass 1200S mass spectrometer [6]. The irradiation parameter (J) for each tray was calculated from an average of the three standard minerals on the tray. An infrared microscope for temperature estimation of each stage of a step heating experiment on a single grain was used.

For site MBE, two hornblende grains were step heated. The spectra show remarkable similarity, with indication of excess argon at the lowest temperature steps followed by argon loss up to 850 °C (Fig. 3). Above this temperature, plateaux are seen in both the age and Ca/K spectra up to about 1050 °C for both grains. The plateau ages for the

TABLE 3

$^{40}Ar/^{39}Ar$ hornblende plateau ages

Sample name	Grain	Number of fractions [a]	Temperature range (°C) [b]	% ^{39}Ar range [c]	Plateau Ca/K with standard error	Plateau age (Ma) with standard error
MBE	1	4	925–1050	17.0 ± 68.0	7.603 ± 0.019	2691 ± 4
	2	6	925–1000	11.0 ± 73.4	6.205 ± 0.009	2683 ± 3
MBC	1	1	900	17.6 ± 75.1	6.025 ± 0.016	2675 ± 4
	2	6	900–1025	6.5 ± 73.4	6.413 ± 0.011	2704 ± 3
Composite [d]		5	900–1000	8.3 ± 71.4	6.399 ± 0.006	2687 ± 3

[a] Number of fractions in plateau.

[b] Temperature range of the plateau.

[c] Percent range of ^{39}Ar from the original spectrum used in calculation of the plateau.

[d] The composite spectrum is made up of a weighted sum of the four individual spectra, "repartitioned" with a synthetic temperature schedule.

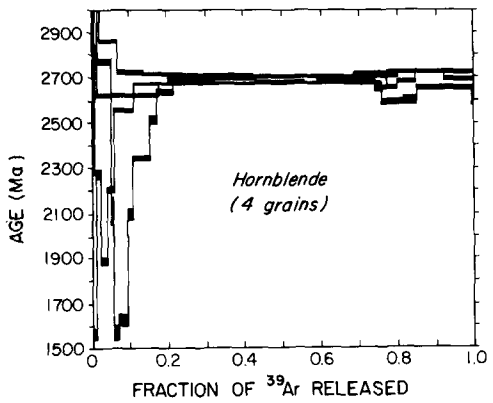


Fig. 3. $^{40}Ar/^{39}Ar$ age versus fraction of ^{39}Ar released for four individual hornblende grains from the Mbabane Pluton; two from site MBE and two from site MBC. The width of each box (which denotes an individual fraction) is $\pm 1\sigma$ in age.

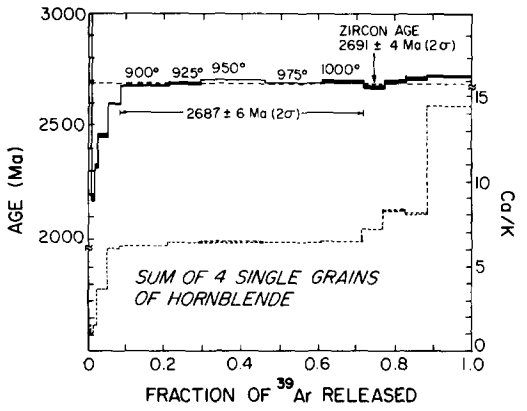


Fig. 4. Composite hornblende age (solid line) and Ca/K (broken line) spectra created by adding the 4 individual spectra (Fig. 3) together and repartitioning to a synthetic heating schedule. The fractions used in the calculation of the plateau age are labelled with their temperatures. The width of each box (which denotes an individual fraction) is $\pm 1\sigma$ in age. The $^{207}Pb/^{206}Pb$ zircon age is shown for reference.

two grains are 2691 ± 8 Ma (2σ) and 2683 ± 6 Ma (Table 3). Above 1050 °C there is a change in the Ca/K ratio. We believe that only the region from 850° to 1050 °C reflects the "true" hornblende portion of the mineral.

The spectra from two hornblende grains from site MBC are slightly different to those from MBE. In these spectra the low-temperature fractions show indication of excess argon, and little evidence of argon loss is seen; however, stable plateaux are reached in the temperature region from about 900° to 1025 °C (Fig. 3, Table 3). These plateaux are at 2675 ± 8 Ma and 2704 ± 6 Ma.

The spectra from the four grains were added together to form a single composite spectrum (Fig. 4, Table 2). The method used in combining the spectra consists of constructing argon volume release curves (versus temperature) for each isotope and then reading off the volume released for each temperature interval of "synthetic" heating schedule. The composite spectrum, then, is a sum of the four individual spectra, weighted by each grain's gas volume. This composite preserves all the key features of the original four spectra. An age plateau from 900 to 1000 °C (63% of the ^{39}Ar release), which also has a constant Ca/K ratio of 6.40 ± 0.01, has an age of 2687 ± 6 Ma (2σ) (Table 3), and we feel that this plateau age is a better indication of the mean hornblende age than the plateau from any single grain. The fact that the mean hornblende age from four grains is indistinguishable from the mean zircon age of 2691 ± 4 Ma implies that the mineral systems closed, for argon and lead diffusion respectively, essentially at the same time.

Five biotite grains were step heated. One grain was from site MBE, one from site MBC while three of these grains were obtained from a crush of material made up of samples from several sites (MBB, MBC, and MBE). No consistent age emerges from the spectra from these five grains (Fig. 5, Table 2). Integrated ages vary from 2088

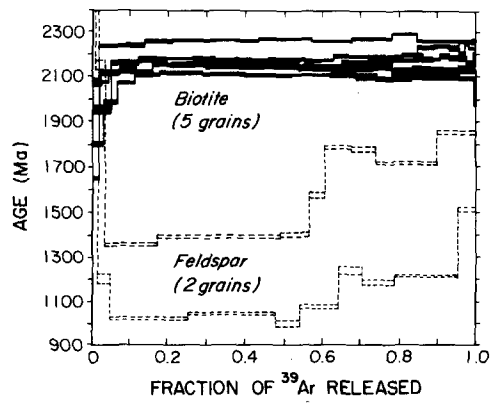


Fig. 5. $^{40}Ar/^{39}Ar$ age spectra from five individual biotite grains (solid lines) and two feldspar grains (broken lines). The width of each box (which denotes an individual fraction) is $\pm 1\sigma$ in age.

± 6 Ma (2σ) to 2261 ± 6 Ma, while plateau ages vary from 2090 to 2284 Ma.

Two feldspar grains from site MBE were step heated. The spectra, with integrated ages of 1671 ± 6 Ma and 1230 ± 6 Ma, are very disturbed (Fig. 5, Table 2). At the highest-temperature steps, ages of 1839 Ma and 1498 Ma respectively are reached. Due to the low activation energy of feldspar for

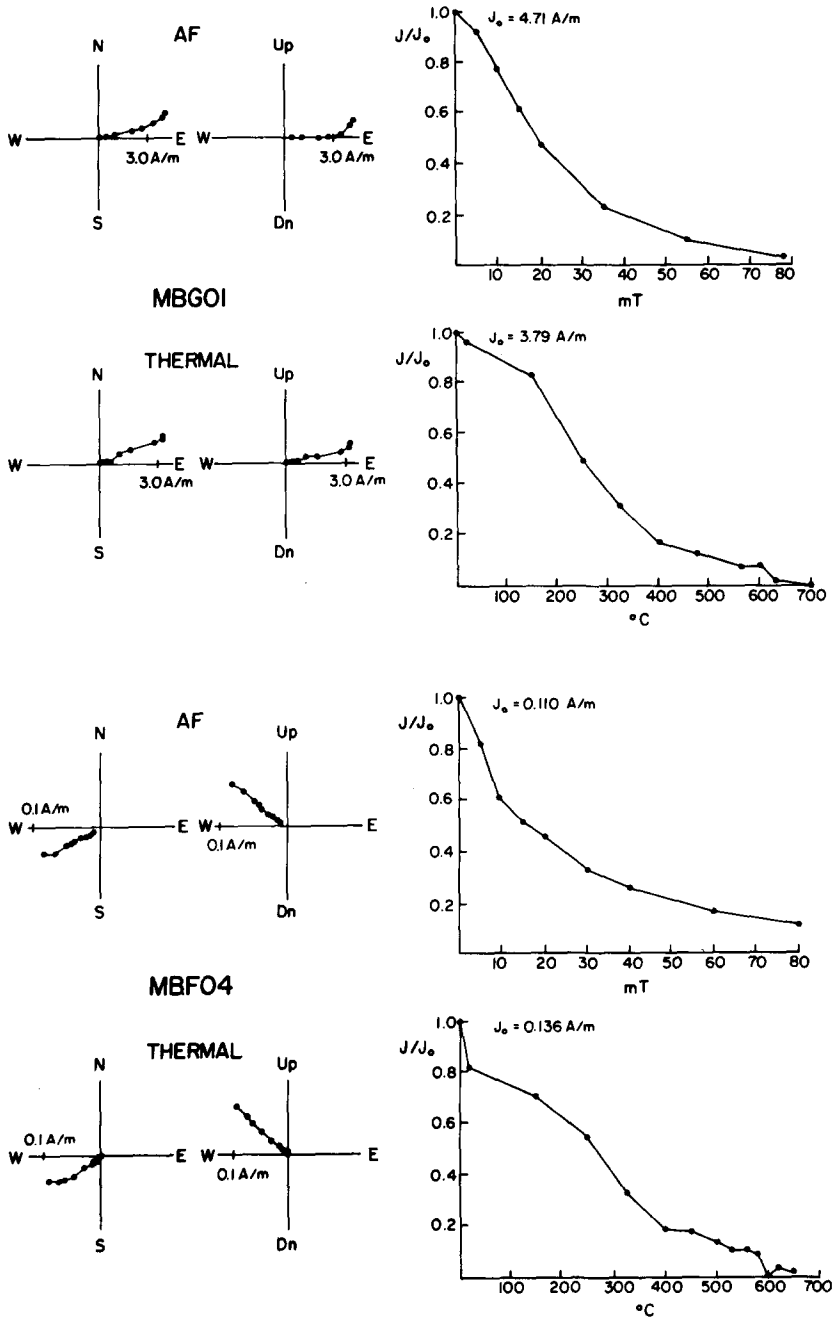


Fig. 6. Orthogonal vector plots and normalized intensity versus demagnetization step for AF and thermal demagnetizations of representative samples from sites MBG and MBF in the Mbabane Pluton. The AF and thermal experiments were done on adjacent specimens from the same sample core. Stable, antipodal magnetization directions are seen in these two sites.

argon diffusion, it is probable that these grains were open to argon diffusion for much of the Proterozoic, perhaps due to shallow burial. We do not attach any significant meaning to the ages.

6. Discussion of the age of the pluton

The remarkable similarity between the zircon and hornblende ages suggests that the Mbabane Pluton underwent rapid cooling, at least through the hornblende blocking temperature, near the time of its intrusion at 2687 ± 6 Ma (using the hornblende plateau age). The scatter in the biotite ages implies that either there was a single low temperature overprinting event at approximately 2000 Ma with each grain recording different amounts of argon loss, or there were several low temperature events distributed over a protracted time interval from 2000 to 2300 Ma, possibly related to dike swarm intrusion.

7. Paleomagnetic study

Stepwise AF demagnetization was performed on one specimen from each sample from the seven sites. Ten specimens from selected samples were stepwise thermally demagnetized up to 650 °C (Fig. 6). The thermal demagnetization experiments generally confirm the magnetic components, isolated from the AF demagnetization, although they better resolve two magnetic components as can be seen by the thermal demagnetization curves (Fig. 6). The higher blocking temperature component shows a direction which is consistently seen in several sites and is isolated in the highest AF demagnetization steps. The lower unblocking temperature or AF coercivity component does not seem to have a consistent direction either within a site or between sites. Stable components were isolated from the highest AF fields or temperatures for all demagnetized specimens using a factor fitting technique [26].

Six sites show within-site consistency of the higher blocking temperature component with a majority of specimens yielding a characteristic northeast shallow direction from which a site mean direction has been calculated (Table 4). The directions for the remaining specimens from the site show no consistent grouping and they are not included in the final analysis. Five sites, MBB, MBC, MBE, MBF and MBG have consistent site mean directions with overlapping circles of 95% confidence. A sixth site, MBA, has a stable mean direction that was displaced from the other five sites. In site MBF, 100% (4 out of 4) of the specimens have directions antipodal to those seen in the other sites, while in sites MBB and MBC, 43% (3 out of 7) were reserved at each site. Of the

TABLE 4

Paleomagnetic results from the Mbabane Pluton

Site	N	AF demagnetization results				Site VGP	
		D (°)	I (°)	κ	α_{95} (°)	lat. (°N)	long. (°E)
MBA	8	21.5	−12.8	11	18	61.5	80.8
MBB [a]	7	73.1	11.8	11	19	12.3	108.0
MBC [a]	7	54.4	−7.0	5	28	33.2	107.1
MBE	5	58.5	12.9	9	27	24.5	99.7
MBF [a,b]	4	66.6	13.2	15	25	17.6	104.1
MBF [b]	4	341.6	−63.5	56	12	66.1	244.5
MBG	7	74.6	13.0	10	20	10.7	108.2
MB mean [c]	6	58.5	5.5	13	19		
MB mean [d]	5	65.4	8.9	43	12		

[a] Site mean has antipodal directions which were reflected through the origin to the northeast quadrant.

[b] Site MBF had two groupings of directions; one of which was consistent with the other sites and one which is parallel to present field.

[c] Mean of all stable sites.

[d] Mean excluding site MBA.

Paleomagnetic pole for the Mbabane Pluton (calculated from site mean VGPs): latitude = 19.7 °N, longitude = 105.7 °E, $A_{95} = 9.4°$ ($N = 5$; Sites MBB, MBC, MBE, MBF, and MBG: $K = 67$).

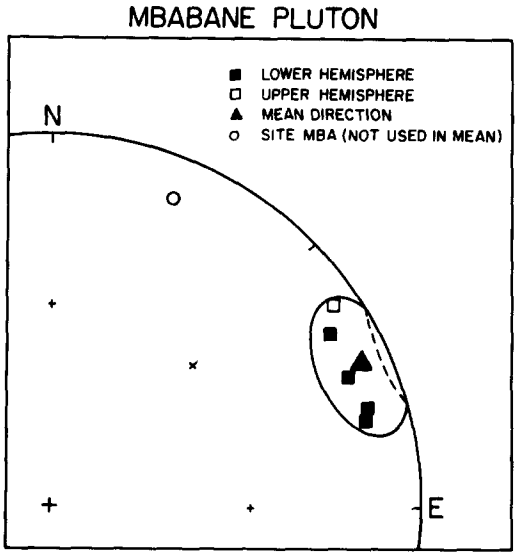


Fig. 7. Equal area stereographic plot of site mean directions from the Mbabane Pluton. Solid (open) symbols are projections on the lower (upper) hemisphere. The large triangle and its α_{95} confidence circle represent the mean direction of the five stable, consistent, site mean directions (excluding site MBA).

30 specimen directions that were used in the calculation of the mean direction (excluding MBA), 10 directions were reversed (33%). Site MBF is slightly more weathered in outcrop than the other sites, and specimens from that site have average NRM intensities one order of magnitude lower than the other sites. Four specimens from site MBF show only a present field direction (VGP at 66.1° N, 244.5° E), four show a shallow direction consistent with the other sites, and two are unstable. Specimens from site MBD do not show site consistent demagnetization vectors, so no site mean could be determined from them.

Table 4 and Fig. 7 summarize the paleomagnetic data. Paleomagnetic poles are calculated for the five consistent sites and for six sites including MBA. When MBA is included, the dispersion increases greatly, and we feel that the paleomagnetic pole calculated excluding MBA is a more representative pole for the Mbabane Pluton. We do not know why the direction from site MBA is rotated 45° counter clockwise relative to the mean from the other 5 sites. One possible explanation is that site MBA has been rotated relative to the other sites after the acquisition of magnetization. Unfortunately, the sampling sites were not mapped accurately enough to confirm this hypothesis.

In general, there appear to be two types of magnetic carriers responsible for the high and low temperature magnetizations. During demagnetization a low unblocking temperature component was removed by 400° C. This may be multidomain or pseudo-single domain magnetite and is responsible for the characteristic shape of the AF demagnetization curves, although tests to identify this carrier were not done. Following the removal of this component, a stable high unblocking temperature component was seen. This component has a laboratory unblocking temperature of 570–610° C (errors in temperature may be ±20° C) and is probably single domain magnetite. All specimens lose up to 90% of their remanence by 600° C.

8. Discussion of the paleomagnetic results

The paleomagnetic pole obtained for the Mbabane Pluton (19.7° N, 105.7° E, $A_{95} = 9.4°$, Fig. 8) is similar to that from the Gaborone Granite [27], located on the western edge of the Kaapvaal Craton in Botswana (Fig. 1). The age of the Gaborone Granite is not well known [22], and Rb-Sr and K-Ar ages range from 2700 Ma to 2150 Ma, similar to the range seen in the Mbabane minerals. It is possible that both poles reflect the same age. However, much more dating work, perhaps on single grains, is needed to resolve the age and thermal history of the Gaborone Granite. If

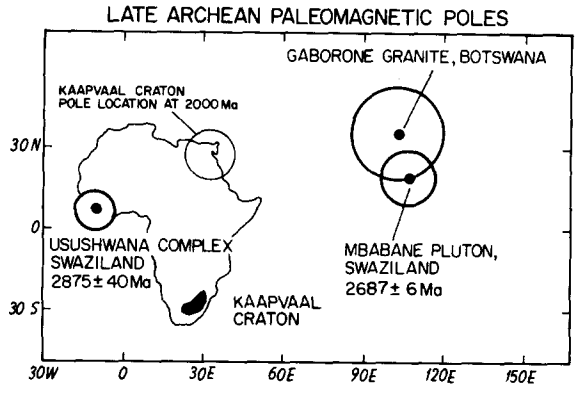


Fig. 8. Mercator projection showing late Archean paleomagnetic poles for the Kaapvaal Craton; the Usushwana Complex pole [18], the Gaborone Granite pole [26] and the Mbabane Pluton pole (this study). The general location of the Kaapvaal Craton paleomagnetic pole at ~ 2000 Ma is shown for reference.

the poles are the same age (~ 2700 Ma), then this would imply that the Kaapvaal Craton had grown into single, stable unit over 500 km wide by the end of the Archean.

The Mbabane Pluton paleomagnetic pole is significantly different from that of the nearby Usushwana Complex [19] which has an age of 2875 ± 40 Ma (Fig. 8).

9. Summary

Either the magnetization seen in the Mbabane Pluton is related to the age of the zircon and hornblende at 2693 ± 6 Ma and therefore the paleomagnetic pole is primary, acquired during initial rapid cooling, or it may be related to the overprinted ages of the biotites and the paleomagnetic pole is a secondary one. We prefer the former interpretation for several reasons.

(1) A magnetic remanence is carried in a phase with a high laboratory unblocking temperature. The magnetic direction associated with this carrier is consistent and can be seen during thermal demagnetization from 400° to 580° after the removal of a low temperature component that does not show a consistent direction. If the magnetization is a TRM, it would be difficult to totally reset the magnetization of this magnetic carrier without disturbing the hornblende argon spectrum. Although there is evidence of slight disturbance of the hornblende grains, as is shown by young ages in the low Ca phases at low laboratory temperatures, there is no evidence of partial resetting of the "true" hornblende phase of the grains.

(2) If the biotite was partially reset by a single event at about 2000 Ma, then it might be expected that a paleomagnetic pole reflecting magnetic overprinting would be seen. Such a pole would be parallel to other poles of that age from the central Kaapvaal Craton (for example, the 1945 ± 40 Ma Witwatersrand Basin overprint pole [28]). The Mbabane Pluton paleomagnetic pole is over 40° away from this direction, and evidence of relative rotation between the pluton and the central Kaapvaal Craton is lacking.

(3) If the argon systems in the biotite grains were reset over a protracted period of over 200 Ma, then it is possible that the magnetization was reset over this period. Given the dispersion in biotite ages, it is unlikely that all sites would magnetically block at exactly the same time, yet the high unblocking temperature component is a single, stable, magnetic direction seen in 5 of 7 sites. Since it is unlikely that all sites would have been remagnetized at the same time, the only way this could occur is if there was no apparent polar wander of the Kaapvaal Craton during this time. Previous work in the Kaapvaal Craton spanning the time from 1800 Ma to 2250 Ma has shown that there has been significant apparent polar wander [29,30]. Thus, it is unlikely that the event(s) which reset the biotite ages has reset the primary magnetization. It is, however, possible that the scattered, low unblocking temperature directions seen in some sites may be explained by such a hypothetical prolonged disturbance.

(4) The magnetic direction seen in the Mbabane Pluton is significantly different from that for the Usushwana Complex which is only 12 km away. As there are no obvious large thermal sources for overprinting near the Mbabane pluton, the overprinting would presumably have to be regional. Since the magnetic directions are different in the two bodies and there is no evidence of fault tilting of either body, the differing magnetization in the two bodies could not have been caused by the same regional event. It is interesting to note that indication of a 2000–2300 Ma event can be seen in the $^{40}Ar/^{39}Ar$ spectra of hornblende and pyroxene bulk separates from the Usushwana Complex [19]. These young ages are associated with low calcium phases and may be related to biotite contamination in the hornblende. Thus, it is possible that some low temperature event reset the argon system in the biotite and biotite phases in both the Mbabane Pluton and the Usushwana Complex. However, it does not appear that this event uniformly reset the the magnetization because of the presence of well defined, magnetic directions in the Mbabane Pluton and Usushwana Complex.

(5) The direction seen in the Mbabane Pluton appears as an overprint in some samples from the Usushwana Complex [19]; however, no Usushwana-type directions are seen in the Mbabane Pluton. This superposition of magnetization is consistent with the presumed ages of 2875 ± 40 Ma for the Usushwana Complex and 2693 ± 8 Ma for the Mbabane Pluton and supports the hypothesis that the two poles are Archean and that

the intrusion of the Mbabane Pluton affected the Usushwana Complex as may be expected given the proximity of the bodies.

10. Conclusion

High precision, single grain dating of the Mbabane Pluton has determined the age of the last major Archean tectonic event in the eastern Kaapvaal Craton. Following the intrusion of the Mbabane and related plutons, no other igneous or tectonic activity apparently effected this area until the intrusion of dikes at about 2000 Ma.

The paleomagnetic pole from the Mbabane Pluton is about 65° different from that of the Usushwana Complex, while the sites are only 12 km apart. There are no known faults between the two bodies or rotations of either body which could account for this difference in pole positions [20,21]. In addition, the higher temperature magnetic direction seen in the Mbabane Pluton is also present, with the same direction, as an overprint in Usushwana Complex samples [19], implying no relative rotation between these two bodies since the intrusion of the Mbabane Pluton. These two well dated paleomagnetic poles from Swaziland, then, provide the first direct evidence that there was significant apparent polar wander, and hence cratonal motion of the Kaapvaal Craton, in the late Archean.

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